Transfer Learning on Multifidelity Data

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Neural Networks as Surrogate Models

Once trained, a NN represents a surrogate whose cost is negligible relative to that of an underlying PDE solver.

This suggests that NNs *might be* valuable in ensemble-based computations, such as

- ⋆ data assimilation (e.g., Bayesian inference, ensemble Kalman filter)
- ⋆ uncertainty quantification (e.g., Monte Carlo)
- ⋆ inversion / parameter estimation, i.e., minimization of the mismatch between predictions and observations
- ⋆ optimization, design, etc.

The viability of NN-based surrogates hinges on the relative costs of NN training and an ensemble-based computation.

NNs in Computational Science & Engineering

Where do data for physics-informed learning come from?

- ⋆ Observational data are seldom sufficient
- ⋆ Simulation data: Set of forward (PDE-based) model runs, $\mathcal{S}_1$
- ⋆ Deployment of a NN-based surrogate is justified if C_1, the cost of generating $\mathcal{S}_1$, is smaller than C_2, the cost of generating a corresponding set for ensemble-based computation, $\mathcal{S}_2$.
- ⋆ Since the training cost of NN is considerable, a NN surrogate is practical if $C_1 \ll C_2$.
- ⋆ Challenge: sizes of $\mathcal{S}_1$ and $\mathcal{S}_2$ are often comparable

Monte Carlo and Neural Network Errors

⋆ Goal: estimate $\mathbb{E}[Q]$ of QoI Q from N_{train} solutions on N_{el} grid

⋆ Mean square error (aka average discrepancy) of the estimator:

MC is unbiased, $$\mathcal{E}^2_{\text{MC}} \sim \epsilon^2_{\text{bias}} + \underbrace{\epsilon^2_{\text{samp}}}_{\mathcal{O}(N_{\text{train}}^{-1/2})}, \quad \epsilon^2_{\text{bias}} = \epsilon^2_{\text{disc}}(N_{\text{el}}) \approx 0$$

NN might be biased, $$\mathcal{E}^2_{\text{NN}} \sim \epsilon^2_{\text{bias}} + \epsilon^2_{\text{samp}}, \quad \epsilon^2_{\text{bias}} = \epsilon^2_{\text{disc}} + \epsilon^2_{\text{est}}$$

⋆ Empirical evidence: $\mathcal{E}^2_{\text{NN}} = \mathcal{O}(N^{\alpha}_{\text{train}})$ with $\alpha \in \{-0.35, -0.07\}$; in SciML, α is close to $-1/2$ and often worse. (Propp & Tartakovsky, JMLMC, 2025)

⋆ A path forward: reduce the cost of generating elements of $\mathcal{S}_1$

NN Training on Mutlifidelity Data

How to reduce the cost of generating elements of $\mathcal{S}_1$?

⋆ In the spirit of multilevel Monte Carlo, our $\mathcal{S}_1$ consists of a large number of low-fidelity elements and a few high-fidelity elements:

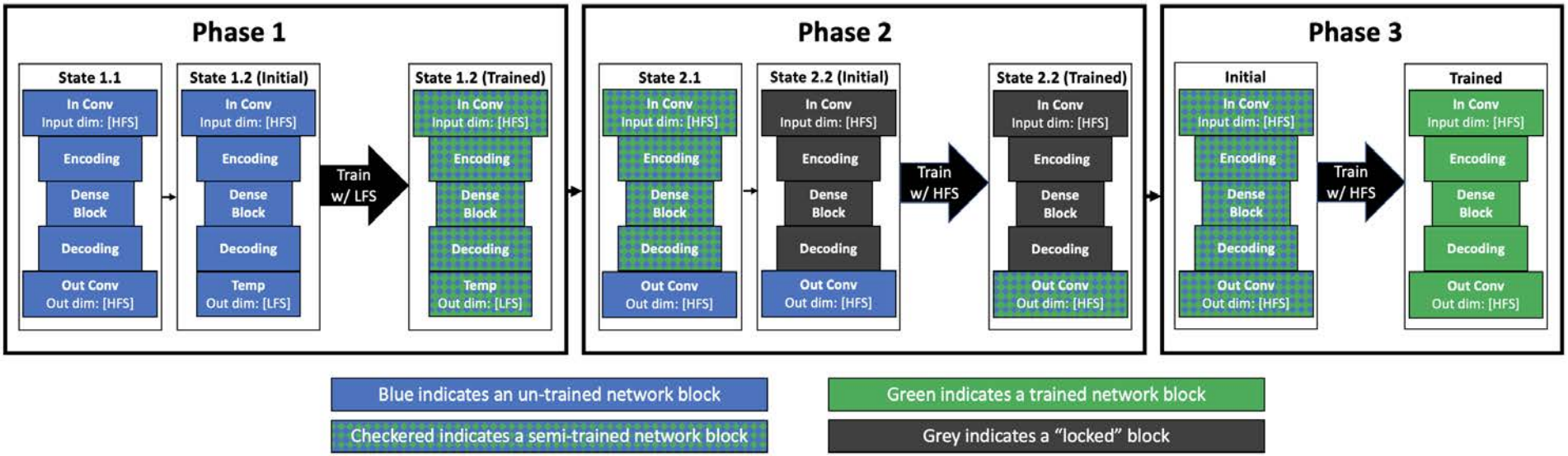
- PDE solutions on coarse and fine meshes
- Alternative models of varying complexity/fidelity

⋆ To be specific, we use a CNN with encoder-decoder architecture. Its weights $\mathbf{w}$ are obtained by minimizing a loss function

$$\mathcal{L}(\mathbf{w}) = \sum_{m=1}^{N_{\text{train}}} \sum_{i=1}^{N_{\text{el}}} \sum_{k=1}^{N_{\text{ts}}} |\mathbf{u}(\mathbf{x}_i, t_k; \boldsymbol{\theta}_m) - \hat{\mathbf{u}}_{ik}(\mathbf{w}; \boldsymbol{\theta}_m)| + \lambda \sum_{n=1}^{N_{\text{w}}} w_n^2$$

Transfer learning is used to patch low- and high-fidelity NNs.

Transfer Learning Strategy

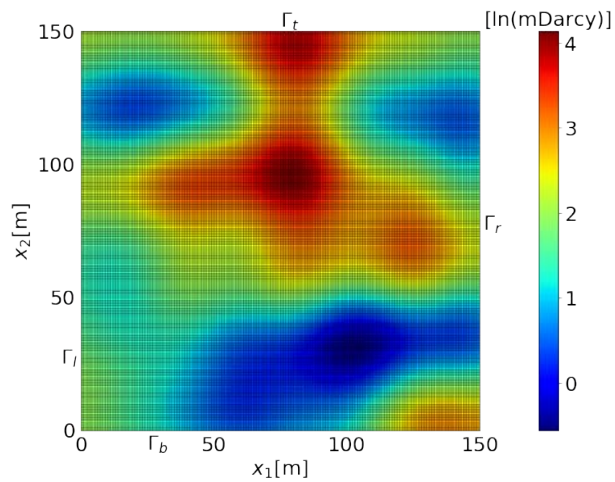


Multiphase flow in heterogeneous media ($\ell = 1, 2$)

$$\phi \frac{\partial S_\ell}{\partial t} = \nabla \cdot (k(\mathbf{x}) k_\mathrm{r} \nabla P_\ell)$$

$$k_\mathrm{r} = k_\mathrm{r}(S_1), \qquad S_\ell = S_\ell(P_\mathrm{cap})$$

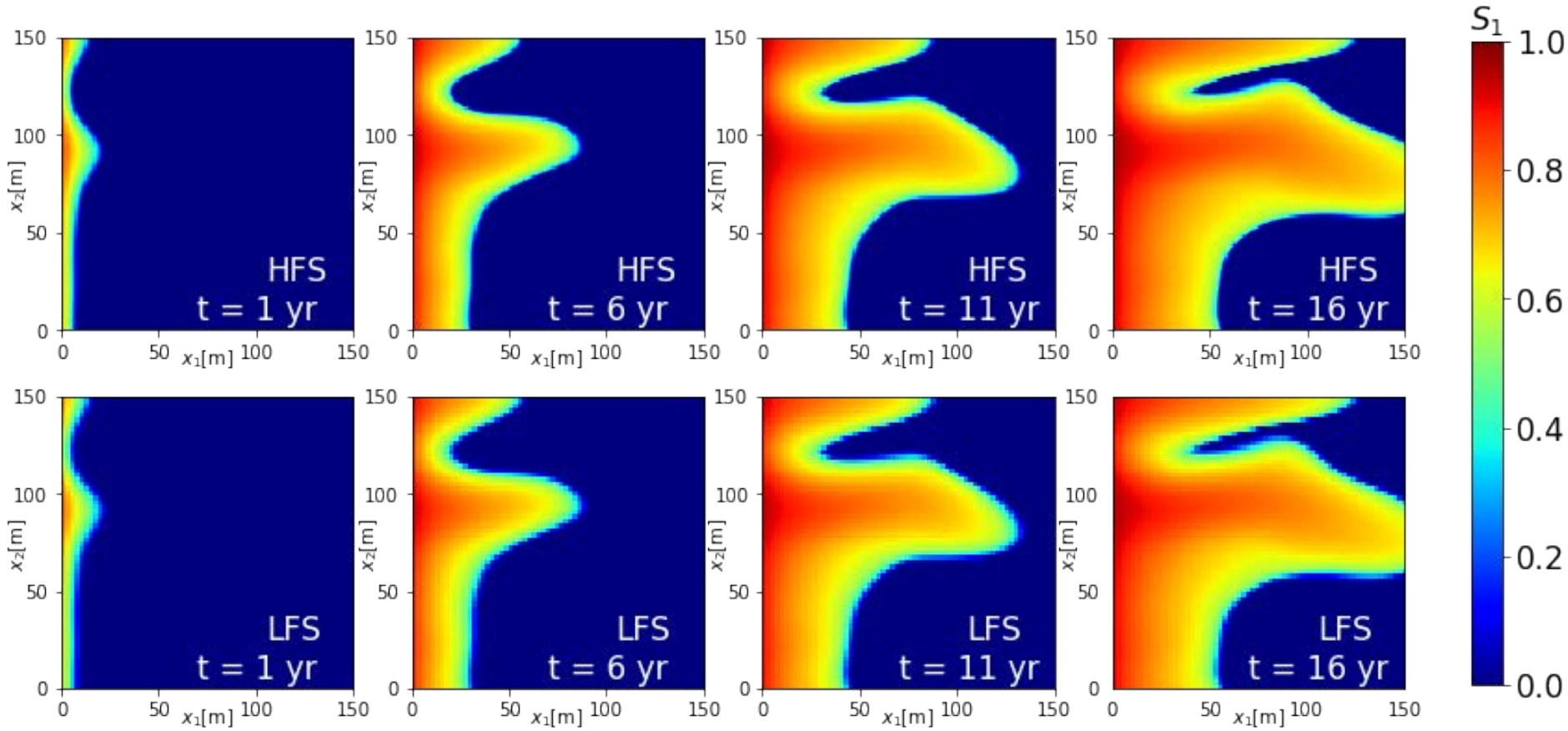
$$S_1 + S_2 = 1, \qquad P_1 - P_2 = P_\mathrm{cap}$$



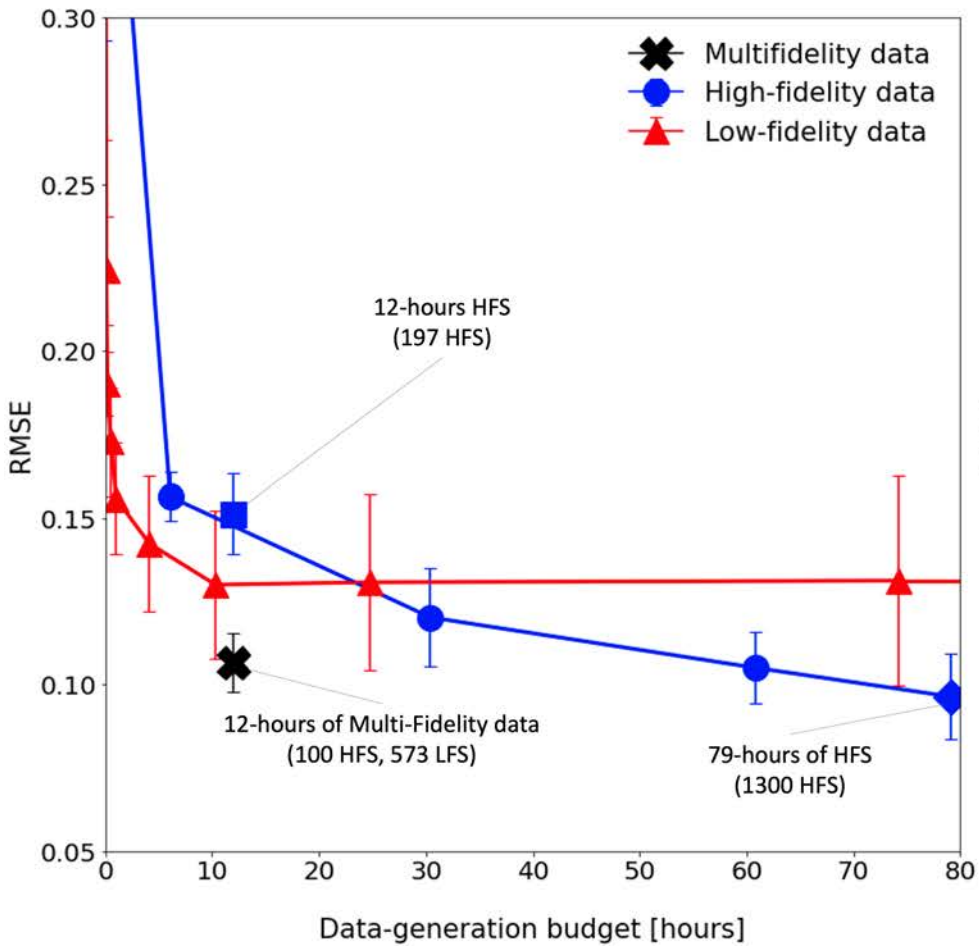
Also, P. Stinis et al.

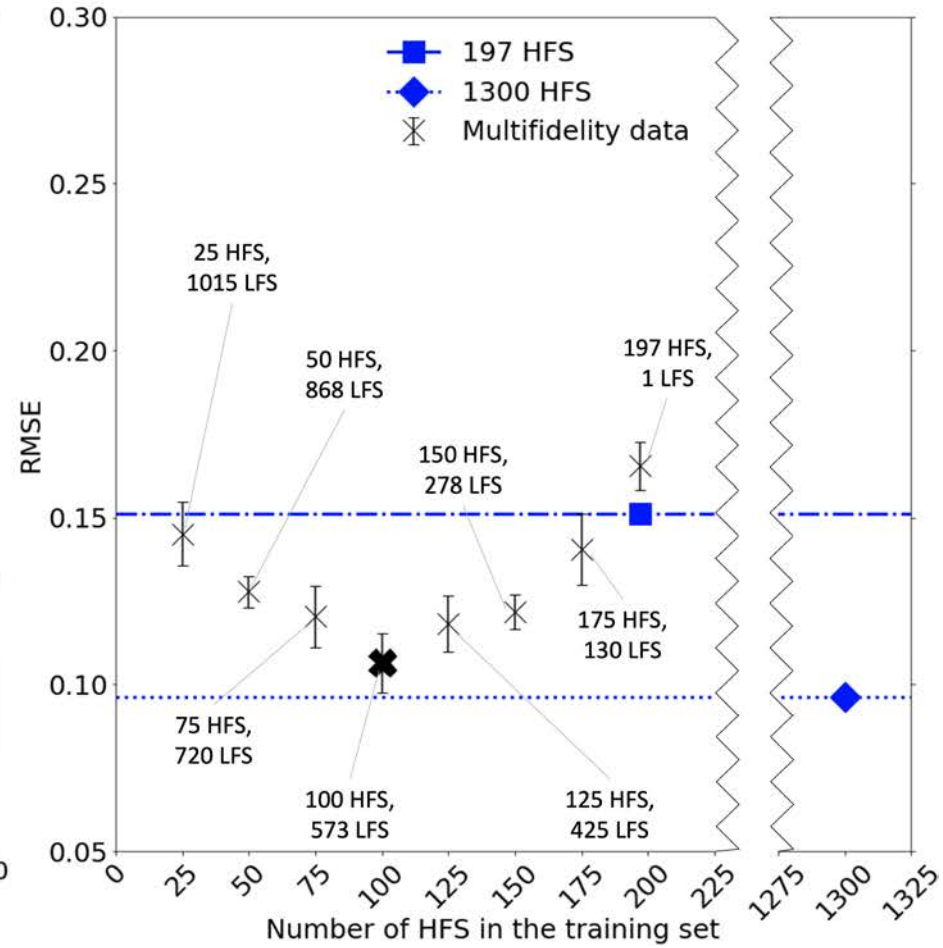
Multiresolution Data

$N_{\text{ts}} = 16$ temporal snapshots of saturation $S_1(\mathbf{x}, t)$ computed on $N_{\text{el}} \times N_{\text{el}}$ grids with $N_{\text{el}} = 128 \equiv N_{\text{el}}^{\text{HFS}}$ and $64 \equiv N_{\text{el}}^{\text{LFS}}$.



Resource-Constrained CNN Training



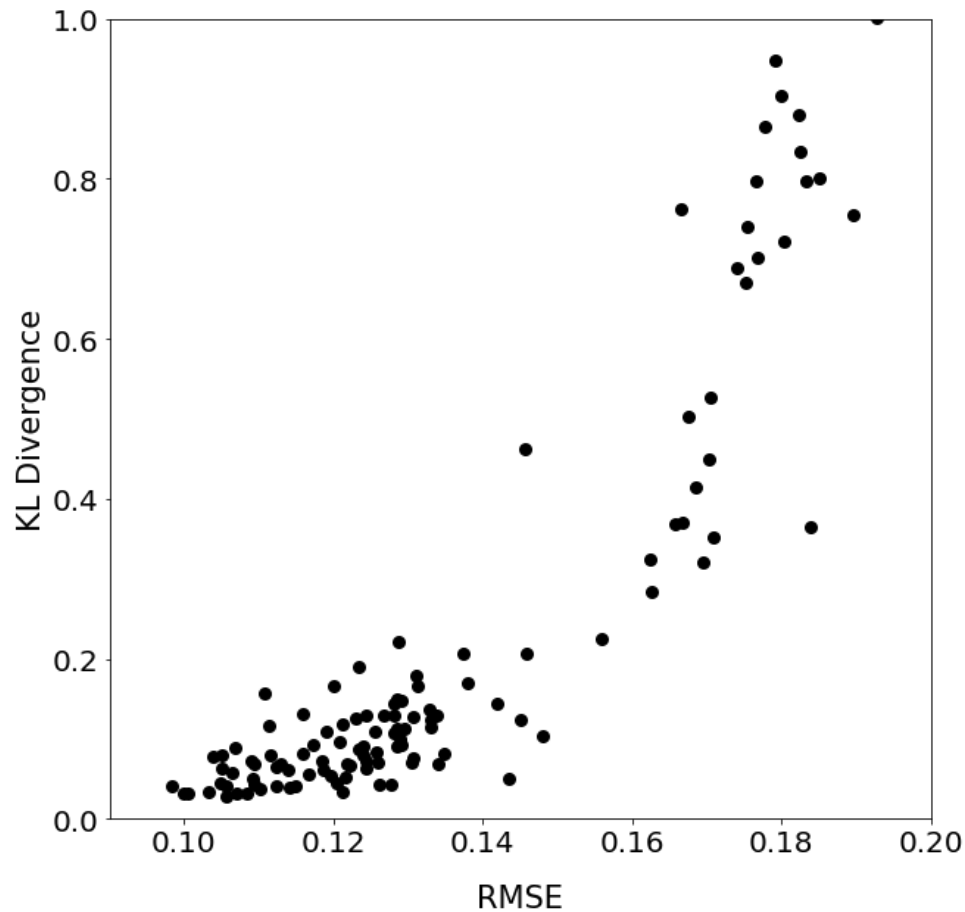


RMSE on test data for the alternative CNN training strategies — similar to resource-constrained ensemble methods for UQ (SIAM/ASA JUQ, 2020; SIAM/ASA JUQ, 2015)

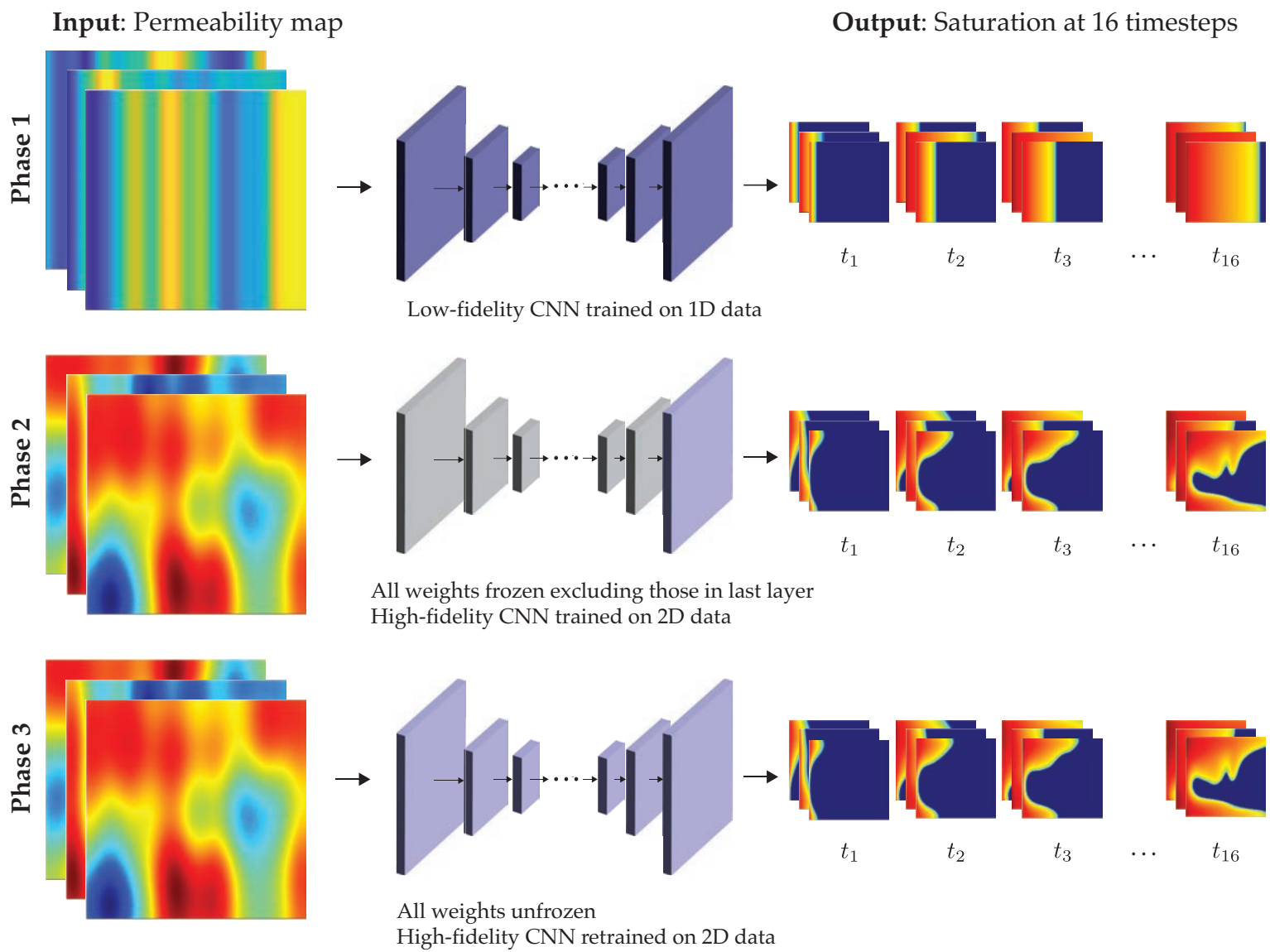
Impact of NN Error on Prediction Accuracy

- ⋆ Quantity of interest: PDF of breakthrough time, f_T
- ⋆ Kullback-Leibler divergence between the "exact" PDF & its NN approximation, f_T^{NN}:

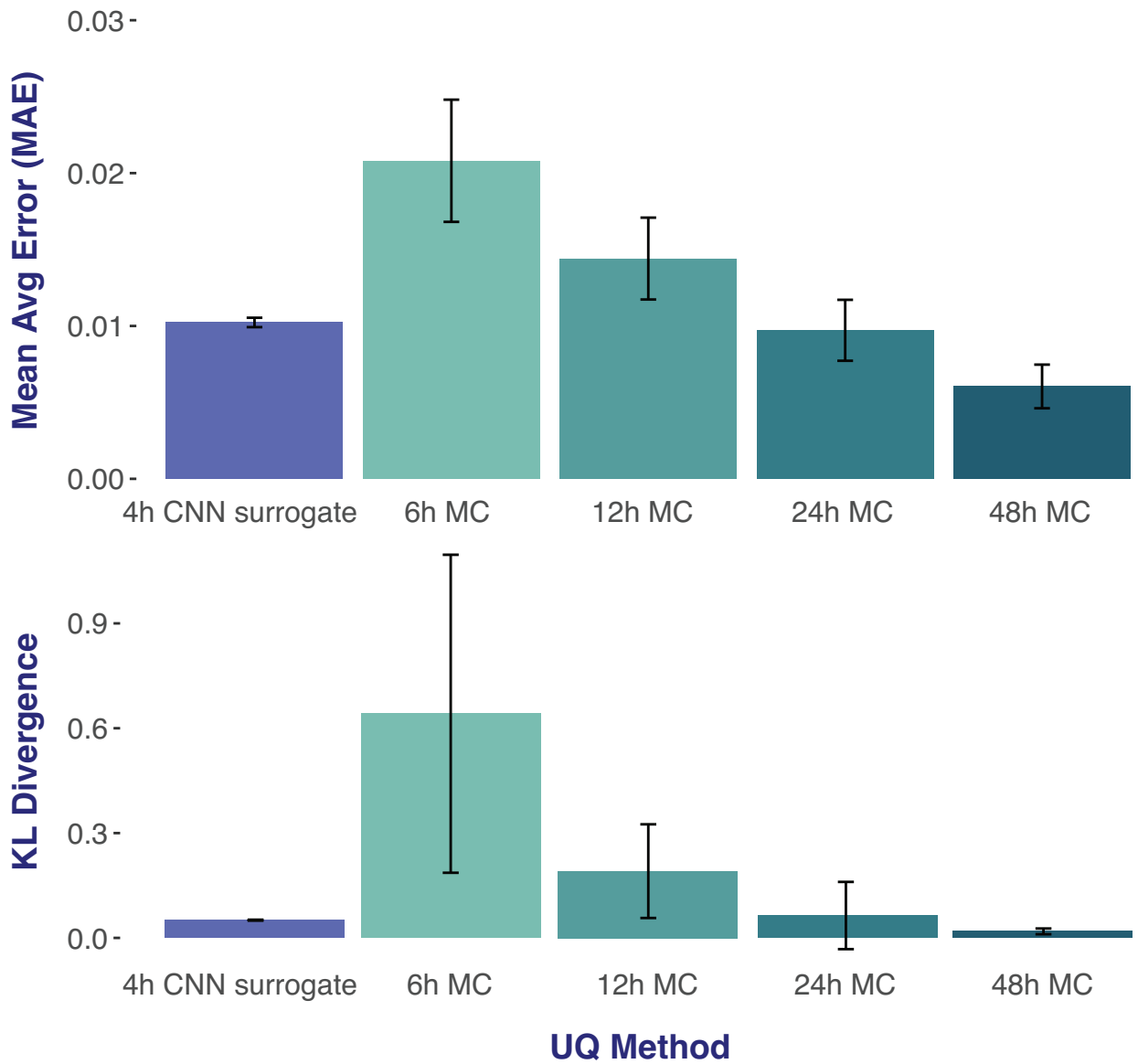
$$D_{\mathsf{KL}}(f_T || f_T^{\mathsf{NN}}) = \int f_T(x) \ln\left[\frac{f_T(x)}{f_T^{\mathsf{NN}}(x)}\right] \mathrm{d}x$$



Multifidelity Data



Relative Performance



Conclusion

- ⋆ CNNs trained on multifidelity data provide an accurate approximation of a fine-mesh PDE solution. A forward pass of the former is two orders of magnitude faster than the latter.
- ⋆ CNN training on multifidelity data reduces the data-generation budget 7-fold relative to training on HFS data. For a small budget, CNN trained on LFS data outperforms its HFS-trained counterpart.
- ⋆ Performance of multifidelity CNNs depends on the ratio between HFS and LFS in the training set. Theory of multilevel Monte Carlo guides selection of an optimal mix of low- and high-fidelity data.
- ⋆ CNN-derived estimates of the PDF of the QoI are close to those of converged high-fidelity Monte Carlo; but the former are three orders of magnitude faster to obtain than the latter.

Acknowledgements

⋆ References

D.H. Song and D.M. Tartakovsky, Transfer learning on multi-fidelity data, *J. Mach. Learn. Model. Comput.*, 3(1), 31-47, 2021

A.M. Propp and D.M. Tartakovsky, Transfer learning on multi-dimensional data: novel approach to neural network-based surrogate modeling, *J. Mach. Learn. Model. Comput.*, 6(2), 13-27, 2025

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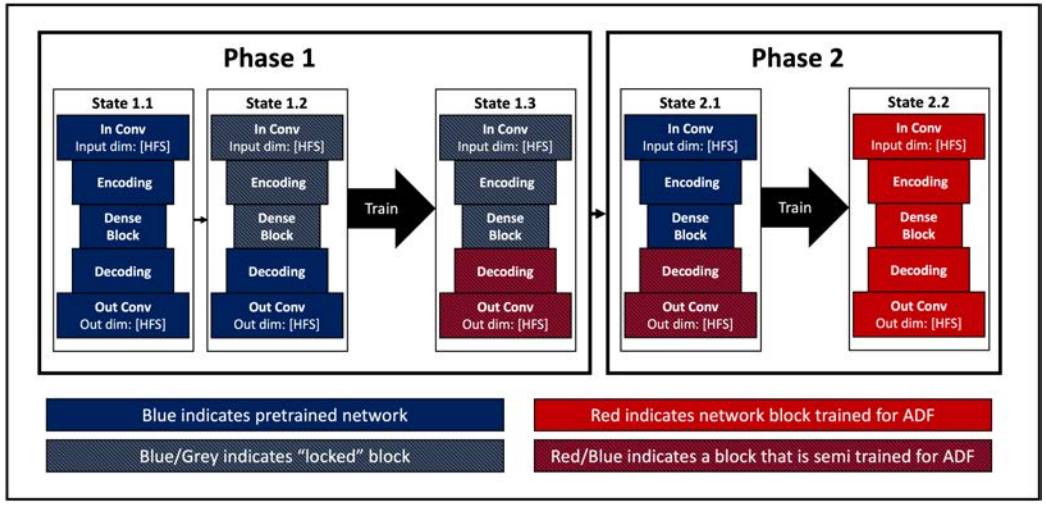
TotalEnergies

Upcycling of Pretrained NNs for New Problems

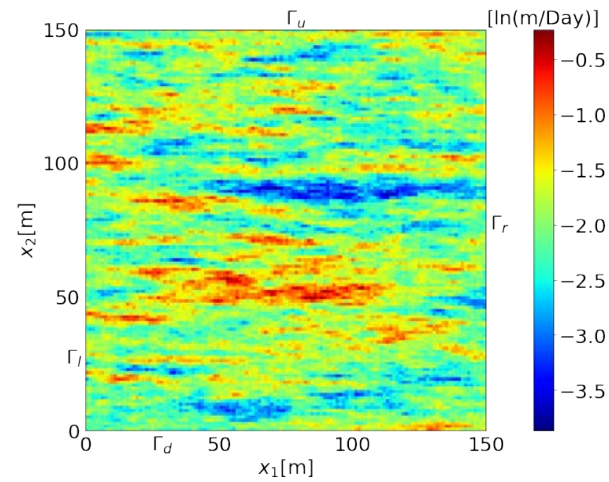
Transfer learning for reuse of pretrained NN surrogates



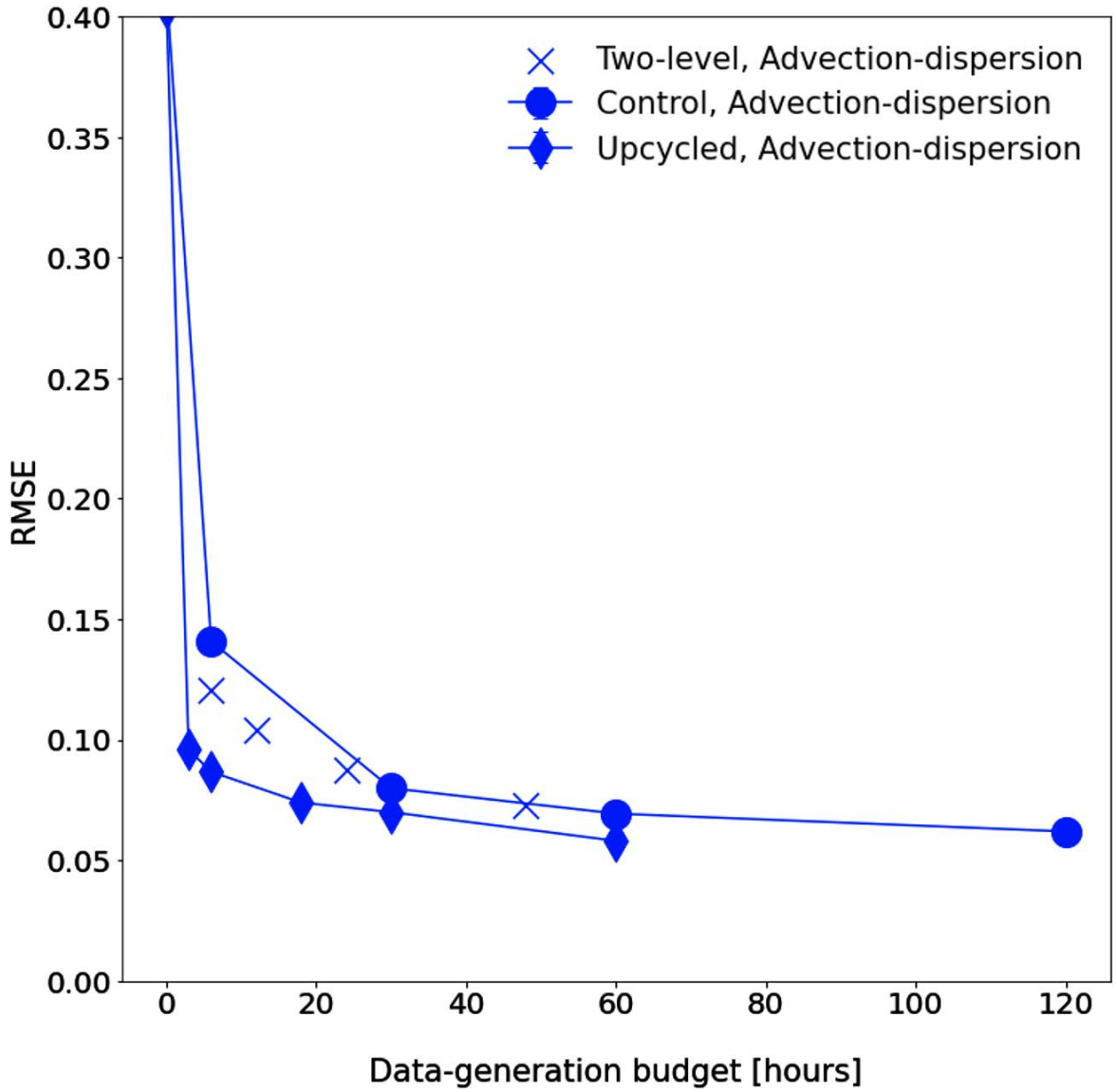
Solute transport in porous media

$$\frac{\partial \phi c}{\partial t} = \nabla \cdot (\mathbf{D} \nabla c) - \nabla \cdot (\mathbf{v} c) + q_s c_s \delta(\mathbf{x} - \mathbf{x}_s)$$

$$\nabla \cdot (K \nabla h) = 0 \qquad \Rightarrow \qquad \mathbf{v} = -\frac{K}{\phi} \nabla h$$



Post-Training Performance



Conclusion for Part II

- ⋆ Starting with a pretrained CNN, rather than a `PyTorch` default-initialized CNN, reduces the data-generation budget five-fold.
- ⋆ For every budget, the pretrained CNN exhibits a significantly smaller RMSE on test data than the default-initialized CNN does.
- ⋆ The use of pretrained CNNs gives outperforms, in terms of required data-generation budget, the the use of multifidelity data.
- ⋆ Disclaimer: The past results are no guarantee of future performance.